AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 2 · STUDENT LESSON

Features, Labels, Targets

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Mark prediction time. Select available inputs. Keep outcomes separate.

Build a feature contract that works for both completed history and a genuinely new request.

Keep one habit: Library snapshot contract → plant-time trace → three complete tables → boundary/missing probes → independent dispatch audit.

MAIN LESSON · SECTION 1

Could this input really exist when the prediction is needed?

> Separate current features, tracking metadata and later observed outcomes.

Your learning goals

I can separate **prediction input**s from observed **target**s.

I can check whether each **feature** exists at **prediction time**.

I can repair **target leakage** and state a **feature contract**.

Recall training/test roles

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from The ML Lifecycle. Give an example and a mistake that the idea helps you avoid.
- After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.
- Why must held out **label**s not be used to choose the threshold in Week 1?

Your end goal

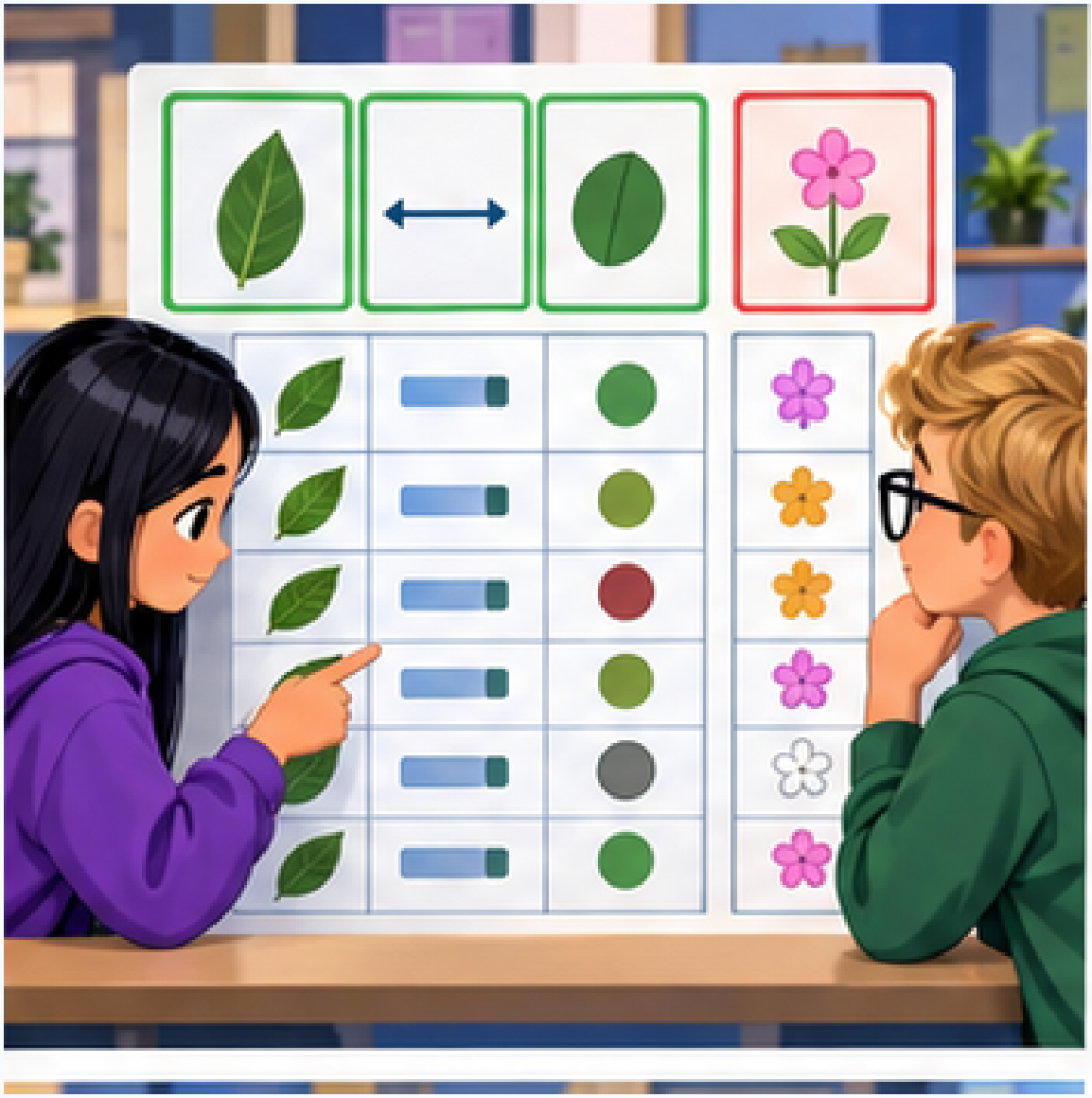
By the end, I can explain **feature**s, **label**s, **target**s using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Features are the information available at **prediction time**; the **target** is the value the model should estimate; **label**s are observed **target** values in examples.



LEARNING **TARGET** A table has leaf length, edge type and verified species. A model predicts species using all three columns. Identify leakage and a legitimate **input** set.

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

Predict lateness at the instant of checkout

> A historical export combines facts from different times.

Read the prediction-time problem

Week 1 separated **training** examples from testing examples. This week asks a different question: could each **input** actually be known when a **prediction** is needed? A historical table may contain facts added after the event.

A model trained with those facts can appear successful even though the **input**s will be missing when it is used.

Use fictional library loans. At the instant a book is checked out, predict whether it will be returned more than 14 days later.

For completed historical loans, **label** 1 means days_to_return is greater than 14; **label** 0 means it is at most 14. Exactly 14 days is on time. Assume no changed due dates in this exercise.

All records and values are invented; do not collect classmates' borrowing histories.

Inspect every field’s meaning and time

Field	When known	Role for this task
pages	At checkout	Candidate feature
queue_at_checkout	At checkout	Candidate feature
days_to_return	After return	Creates the label only
returned_late	After return	Label to predict
loan_id	At checkout	Tracking metadata

Keep the original queue snapshot and arbitrary ID separate

queue_at_checkout is a saved count of waiting requests at that instant. It is not the queue when we later export the table. Candidate means available to consider, not proven useful or appropriate.

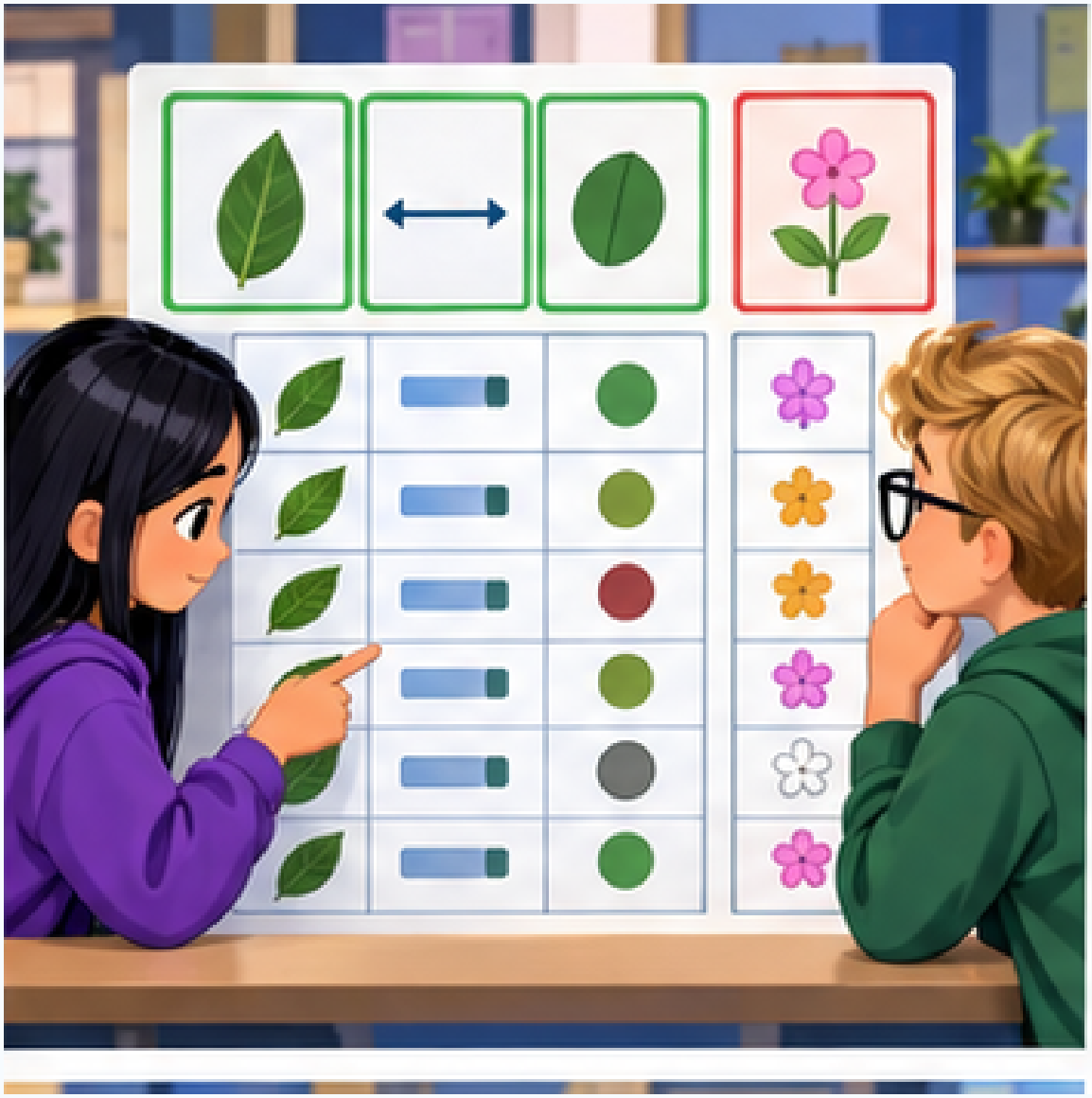
Exclude the arbitrary loan identifier from this exercise; keep it separately to track records.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.
</details>



MYSTERY The team must use **feature**s, **label**s, **target**s to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Use the same allowed fields for past and new records

> A feature list should be reproducible without the future target.

Read the complete ordered feature contract

Library feature contract	Declared rule
Prediction time	Checkout, before the return outcome
Feature order	pages then queue_at_checkout; page count then saved waiting-request count
Values	Positive integer pages; nonnegative integer queue, including 0
Target	returned_late=1 when days_to_return>14, else 0 for completed loans; exactly 14 is on time
Tracking metadata	Arbitrary loan_id is excluded from predictors in this exercise
Excluded inputs	days_to_return, returned_late, fee_after_return and post-checkout renewals
Implementation limit	Feature selection code assumes valid fields and does not fit a model, validate arbitrary records or predict a new outcome

Complete selection function and past/new examples

```
def features(row):
    return [row["pages"], row["queue_at_checkout"]]

past = {"loan_id": "A", "pages": 120,
        "queue_at_checkout": 2, "days_to_return": 18,
        "returned_late": 1}
print(features(past))
print(past["returned_late"])
new = {"loan_id": "B", "pages": 80,
```

```
        "queue_at_checkout": 0}
print(features(new))
```

Trace inputs separately from the historical label

The **feature** order is always pages first, queue count second. The first **output** is [120, 2]; the separate **label** is 1 because this historical loan took 18 days.

The new loan produces [80, 0] even though its return outcome does not yet exist. The same **feature** function serves past and new records.

It deliberately selects allowed fields rather than copying every column except returned_late.

Trace recording time to later learning

Follow information through time

MOMENT	AVAILABLE INFORMATION	ALLOWED USE
Checkout	Pages and saved queue	Prediction inputs
Later return	Elapsed days	Create observed label
Historical training	Inputs plus separate label	Fit a model later

The outcome may teach the model during training, but it cannot be an input to an earlier checkout prediction.

State validation and missing-field limits

This code prepares **input**s; it does not fit a model or predict lateness. It assumes the two **feature** fields contain valid nonnegative integer values and pages is positive. A missing required field raises KeyError.

A real pipeline needs explicit validation and a missing-data policy.

Explain the deciding check

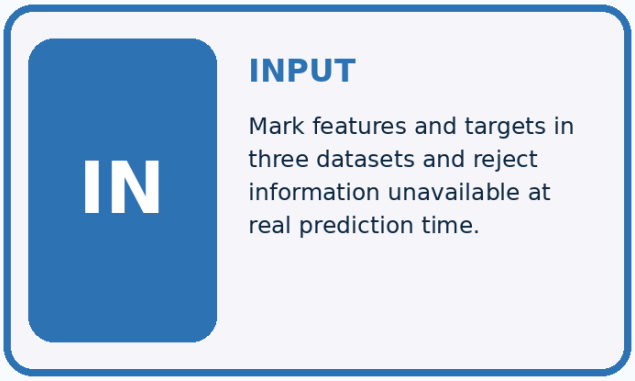
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

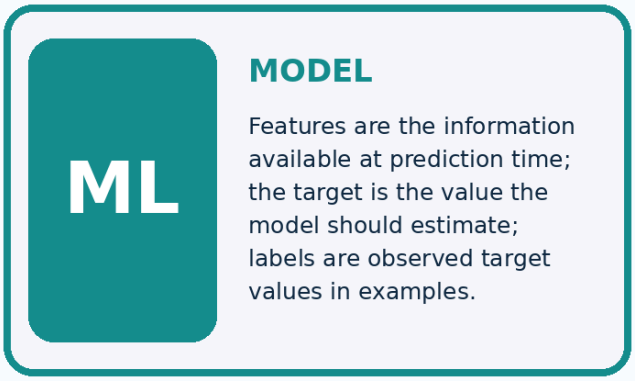
▼ Earlier wording, alternatives and picture transcripts

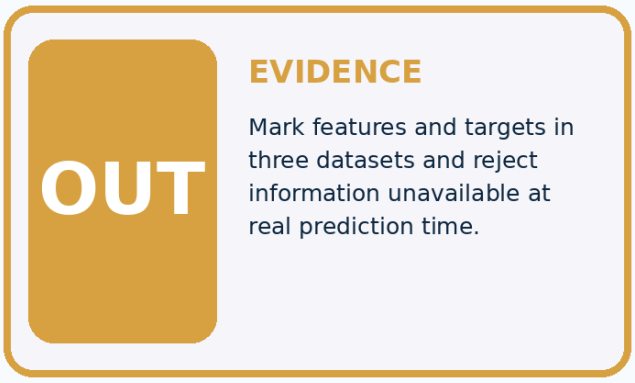
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Mark **feature**s and **target**s in three datasets and reject information unavailable at real **prediction time**.

MODEL IDEA **Feature**s are the information available at **prediction time**; the **target** is the value the model should estimate; **label**s are observed **target** values in examples.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Plant row: leaf_width=6 cm, color_score=0.72, health=spotted. **Feature**s are leaf_width and color_score because they exist before **prediction**.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

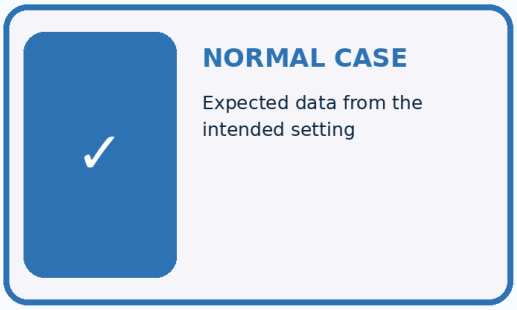
Connect the picture and the explanation

MAIN LESSON · SECTION 4

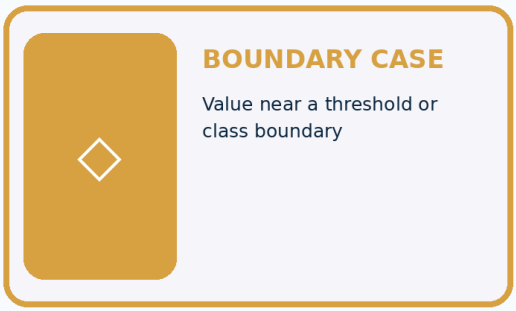
Compare available features with outcome-revealing fields

> An eligible input is a candidate to evaluate, not proof of usefulness or causality.

Four evaluation conditions



NORMAL Expected data and expected behavior.

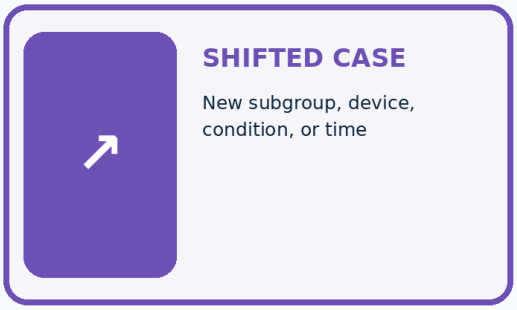


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous **label**, or uncertain **prediction**.



MISSING Necessary field, **label**, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: A table adds tomorrow_soil_moisture as an **input**.

▼ **Check after predicting**

Prediction: Reject it for this **prediction**-time contract.

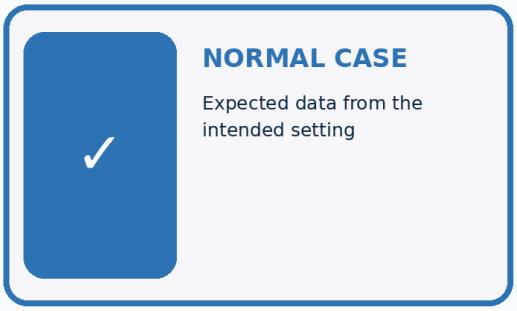
Reason: Tomorrow’s measurement is unavailable when today’s **prediction** must be issued.

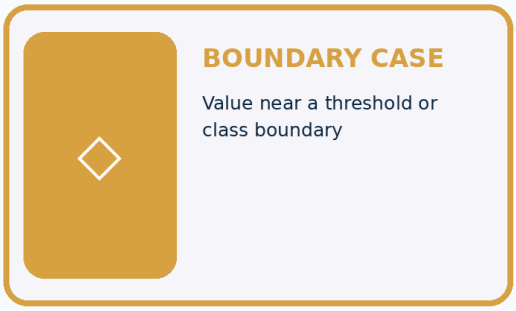
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

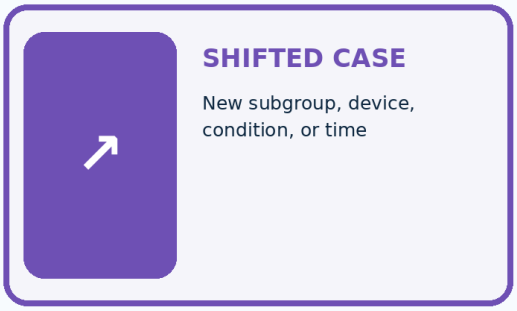
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous **label**, or uncertain **prediction**.

MISSING Necessary field, **label**, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace the prediction-time line in a separate plant task

> Field names alone are insufficient; the recorded meaning and timestamp decide availability.

Read the complete plant task and scope

Choose **feature**s that exist when a plant **prediction** must be made.

Predict next-week wilting at today's photo capture. Available fields: today_light, today_soil_moisture, plant_id; next_week_wilt is the **target** recorded later.

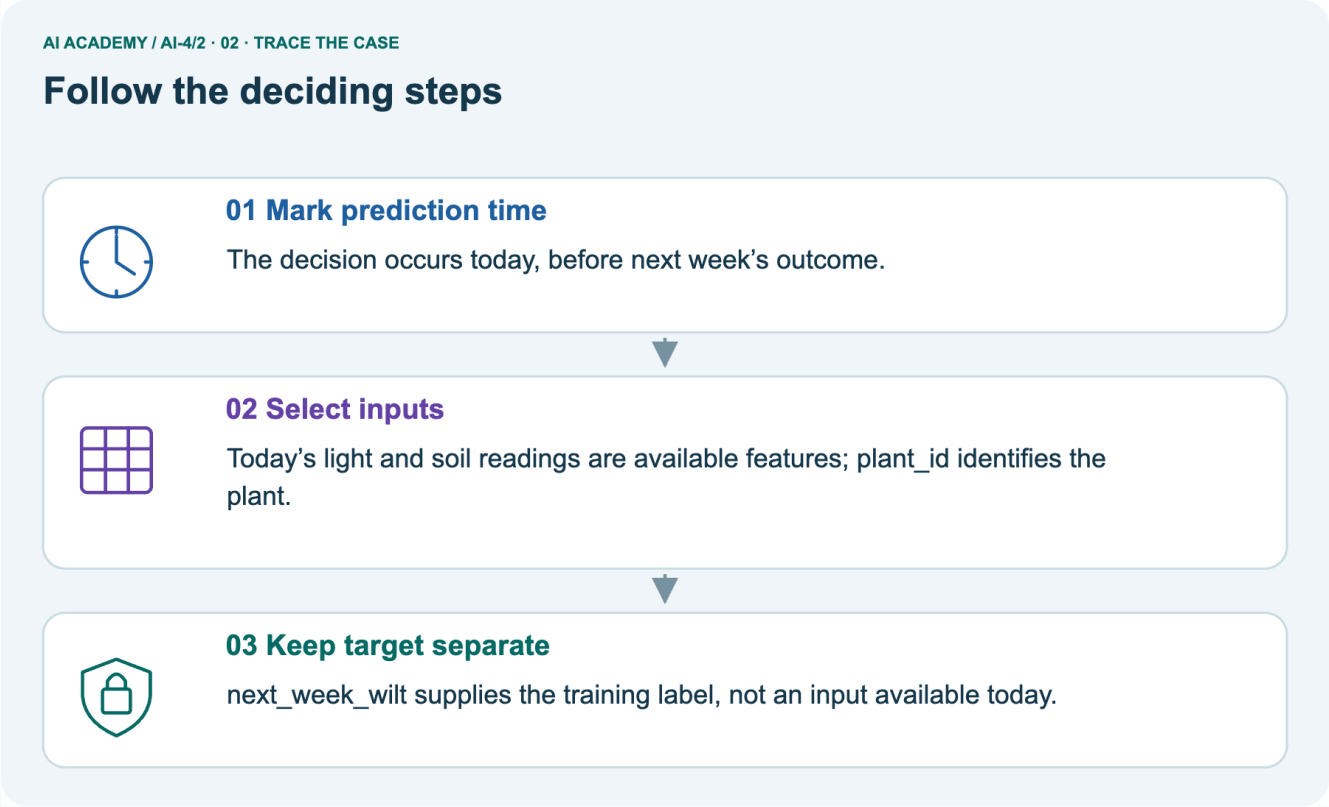
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Separate task contracts. Library lateness is predicted at checkout using pages/queue snapshot. The pictured plant task predicts next-week wilting at today's capture using today_light and today_soil_moisture; plant_id is metadata and next_week_wilt is a later

target. The added tomorrow_soil_moisture field is explicitly an observed reading captured tomorrow, not a forecast available today.

Reject it for today's inputs even if a later export contains it. These are fictional records, not real plant or borrowing data.

Trace available inputs and later label



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Mark prediction time	The decision occurs today, before next week’s outcome.
2. Select inputs	Today’s light and soil readings are available features; plant_id identifies the plant.
3. Keep target separate	next_week_wilt supplies the training label, not an input available today.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

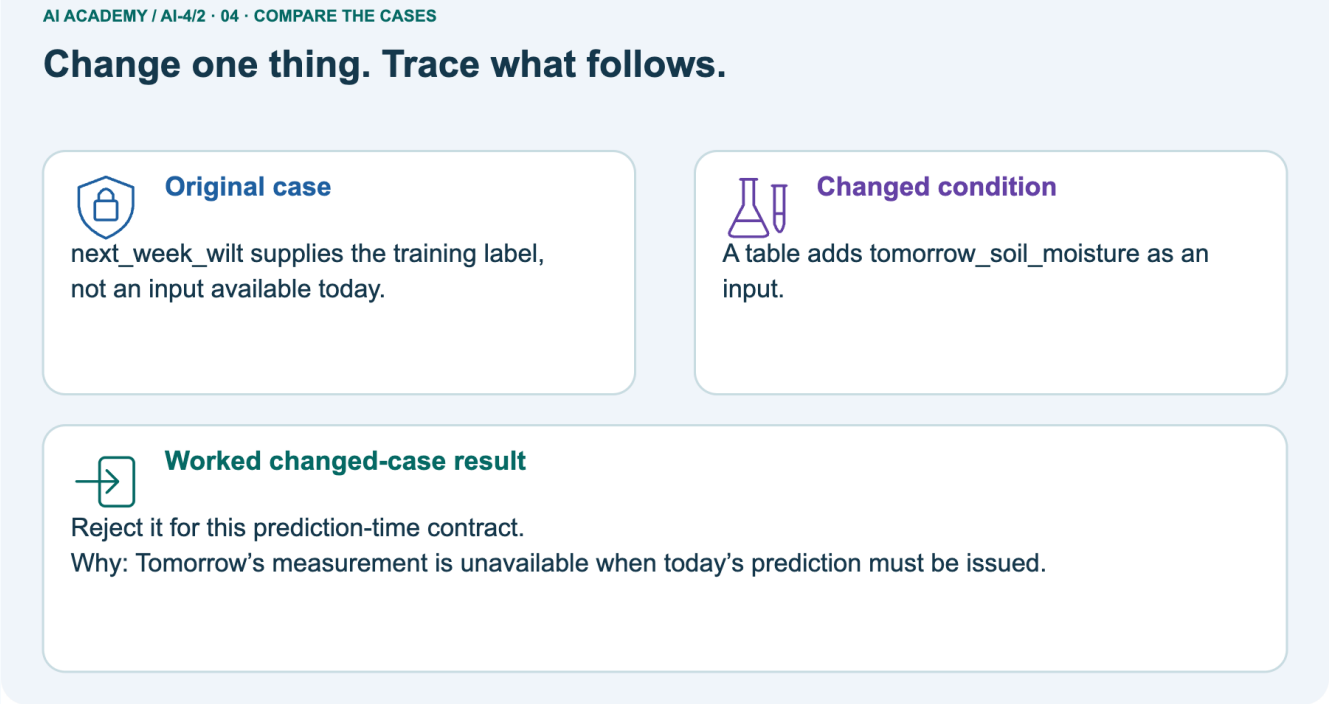
Read the picture in order. next_week_wilt supplies the **training label**, not an **input** available today. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict adding tomorrow’s observed measurement

- Change one thing: A table adds tomorrow_soil_moisture as an **input**. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State usefulness and causal limits

- Eligible fields are not automatically useful or causal. Measure their predictive value with a valid split.

- Read the **label**led picture: Why can a historical table contain a field that is an invalid **input**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

▼ Reference · Library timing questions

For every proposed **input**, ask what it means, when it was recorded, and whether the same definition can be supplied at checkout. A field in the historical export is not automatically available at **prediction time**.

An **input** calculated from later information is also unavailable, even if its name does not mention the outcome.

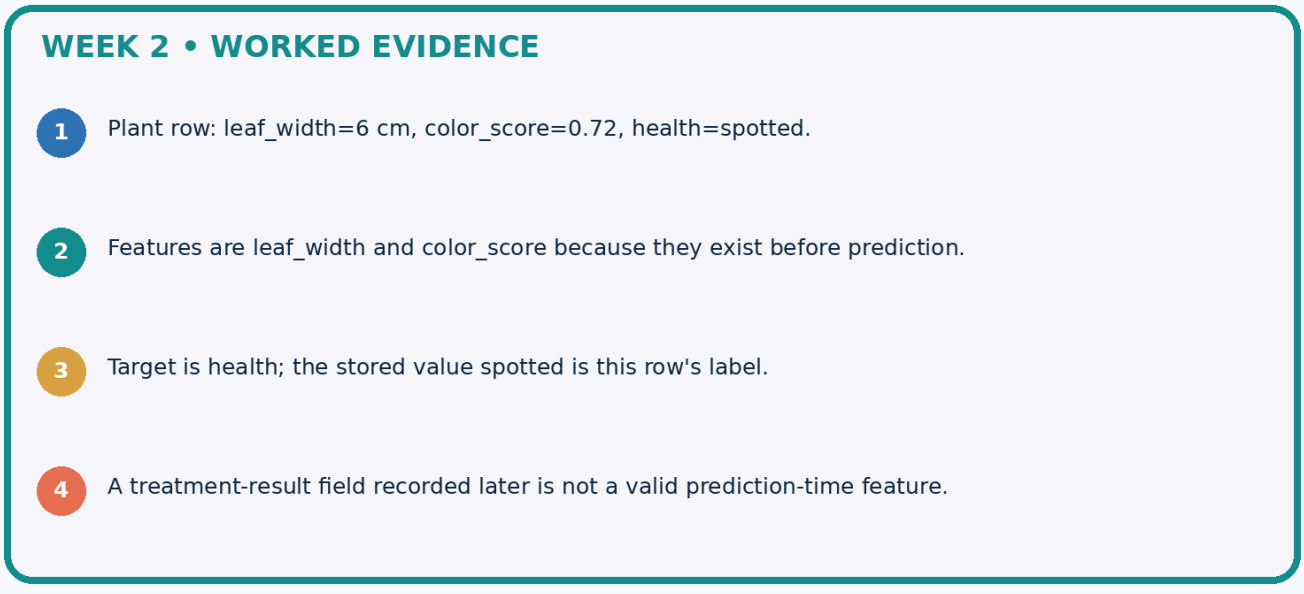
> **A completed loan has pages 200, queue_at_checkout 3 and days_to_return 14. Write the feature list and label. Someone adds overdue_fee calculated after return. Can it become a feature for the checkout prediction? Explain.**

▼ **Earlier wording, alternatives and picture transcripts**

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.



STEP 1 Plant row: leaf_width=6 cm, color_score=0.72, health=spotted.

STEP 2 **Feature**s are leaf_width and color_score because they exist before **prediction**.

STEP 3 **Target** is health; the stored value spotted is this row's **label**.

STEP 4 A treatment-result field recorded later is not a valid **prediction**-time **feature**.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

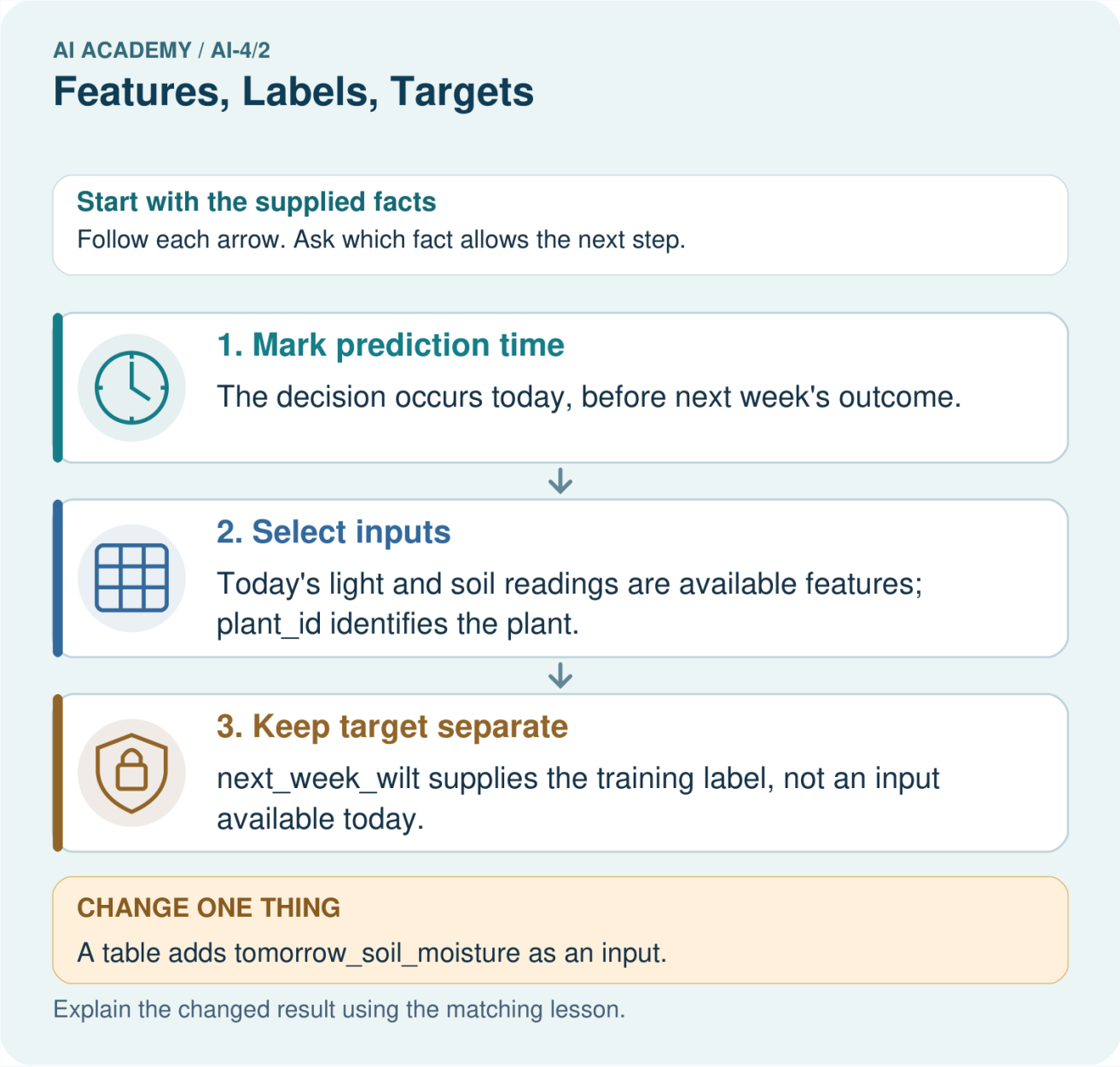
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Features, Labels, Targets

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find the target information hidden in another column

> Dropping the label column alone may leave the answer in the inputs.

Inspect the first-error table

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

State the meaning/timestamp limitation

- A **feature contract** records field names, meaning, units, order and availability time. Here the **input**s are pages in pages, then queue_at_checkout in requests, both captured at checkout.

  The **target** definition is returned more than 14 days later. Historical outcomes may be available for **training**, while new outcomes are unknown.

  Never fill an unknown **label** with 0 merely because the book has not yet been returned.

- Availability alone does not establish predictive value, fairness, a causal effect or a reason to use a model. This lesson audits **input**s; later lessons evaluate models.

  If an **input** cannot be collected at checkout, revise the design or specify a justified missing-**input** policy before use.

Challenge any-stored-column input selection

- Diagnose one cause and name the cleanest next experiment:

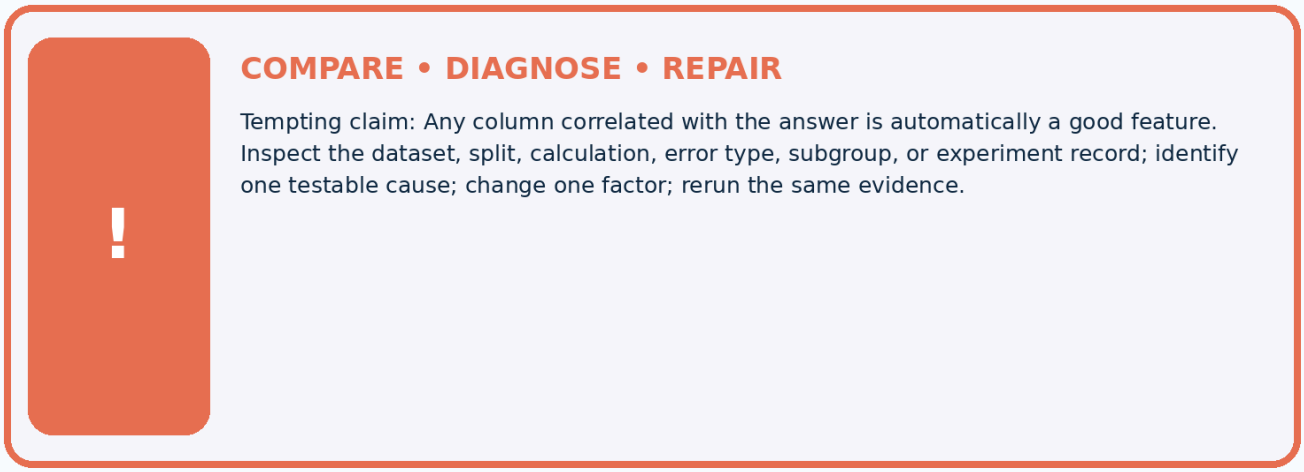
CLAIM TO CHALLENGE Any column correlated with the answer is automatically a good **feature**.

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



A closer explanation

> **Write a two-sentence input contract for this checkout task. Include the feature order and timing, then explain why an unknown future outcome must not be recorded as an on-time return.**

MAIN LESSON · SECTION 8

Practise the 14-versus-15-day label boundary with less help

> The same checkout features can accompany different later outcomes.

Complete library W2

Use workbook W2 for the connected example "**Feature**s **label**s and **prediction time**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Audit stored-column-as-feature reasoning

- Use the worked case for **Feature**s **label**s and **prediction time**. Proposed explanation: "Every stored column is a **feature**". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured later field

- Why can a historical table contain a field that is an invalid **input**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Keep independent dispatch task separate

This is supported practice. Keep the separately named independent case for an attempt before feedback.

Trace direct and disguised target leakage

A learner removes returned_late but keeps days_to_return in the **input**s. That still reveals the answer: compare days_to_return with 14. Remove both the **target** and the later return measurement from **prediction input**s.

Also exclude a fee calculated after return or a renewal made after checkout. These are examples of **target leakage**, where unavailable outcome information enters the predictors.

Moving rows into a test set does not repair this problem if each test row still contains the later return measurement. Rebuild the **input**s to match checkout time and repeat development and evaluation honestly.

If earlier test results influenced changes, obtain a new final evaluation set as well.

▼ **Reference · Reproduction-record fields**

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + **feature**s + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

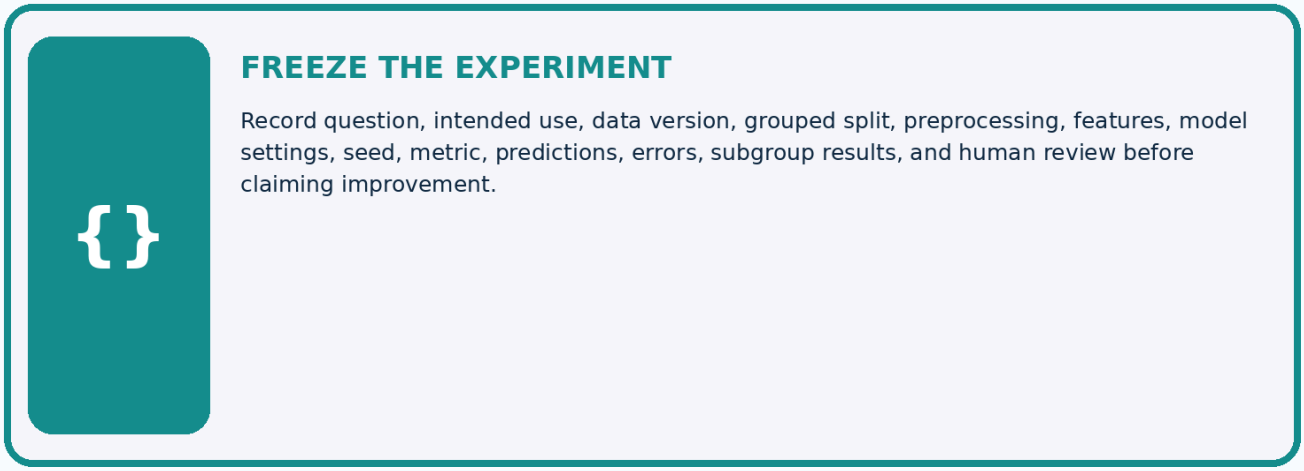
Write the minimum record needed to reproduce today's result:

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

A closer explanation

Practise with less help: Trace the feature contract

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Sort three complete time-aware tables and four probes

> Every field’s example value, recording time and role is supplied.

Table 1 · Library checkout

Library field / complete past L1 value	Captured when	Role at checkout
loan_id=L1	At checkout	Tracking metadata by this exercise’s policy
pages=96	At checkout	Candidate feature 1
queue_at_checkout=4	Saved at checkout	Candidate feature 2
days_to_return=14	After return	Later observed measurement, not input
returned_late=0	Derived after return using >14	Target
fee_after_return=0 toy credits	Assigned after return; 1 per overdue day under fictional tariff	Disguised outcome-derived field, not input

Table 2 · Today's plant capture

Plant field / complete fictional P1 record	Captured when	Role for next-week-wilt task
plant_id=P1	At capture	Tracking/grouping metadata, not input here
today_light=7 invented units	Today	Candidate feature 1
today_soil_moisture=4 invented units	Today	Candidate feature 2
tomorrow_soil_moisture=2 invented units	Observed tomorrow	Not available today
next_week_wilt=1	Next week under supplied observation rule	Target; no real plant condition is being assessed

Table 3 · Puzzle-start prediction

Puzzle field / complete fictional Q1 record	Captured when	Role for slow-completion task
session_id=Q1	At puzzle start	Arbitrary tracking metadata
piece_count=20	At start	Candidate feature 1
clues_at_start=2	At start	Candidate feature 2
finish_minutes=9	After completion	Outcome measurement, not input
slow=1	After completion; 1 when finish_minutes>8	Target; exactly 8 would be 0

Puzzle field / complete fictional Q1 record	Captured when	Role for slow-completion task
award_color=amber	After completion: amber if slow 1, green otherwise	Outcome-derived disguised field, not input

Four complete library probes

Complete library probe	Expected evidence
Normal completed L1: pages 96, queue 4, days 14	Features [96,4], target 0
Boundary revision L1B: same checkout fields, days 15	Features unchanged [96,4], target 1
Missing queue snapshot: pages 96, queue unavailable	Required input unavailable; request data, do not invent 0
Overwritten export queue: current queue 1, original snapshot unknown	Current value cannot stand in for checkout snapshot; preserve missing provenance

Annotate, explain and preserve missing provenance

1. Feature-or-Target Sort: annotate all three complete tables with prediction time, field meaning, candidate features in order, target and exclusions. Find the later-derived fee/award fields rather than removing only the target name. For the library probes, predict feature lists and label changes before the optional code run; code assumes complete valid feature records and does not solve missing snapshots.

   Rebuilding a train/test split alone cannot fix outcome-derived inputs. Keep loan/plant/session IDs as metadata under these declared tasks, without claiming every ID-like field is always useless. No model is fitted and no new outcome is predicted here. Keep the independent parcel-dispatch task withheld.

Complete optional feature/target program

```
def features_at_checkout(row):
    return [row["pages"], row["queue_at_checkout"]]

def completed_loan_target(row):
    return int(row["days_to_return"] > 14)

for days in [14, 15]:
    past = {"pages": 96, "queue_at_checkout": 4, "days_to_return": days}
    print(features_at_checkout(past), completed_loan_target(past))
new = {"pages": 80, "queue_at_checkout": 0}
print(features_at_checkout(new), "target unavailable")
```

Keep the evidence ledger

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____

> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

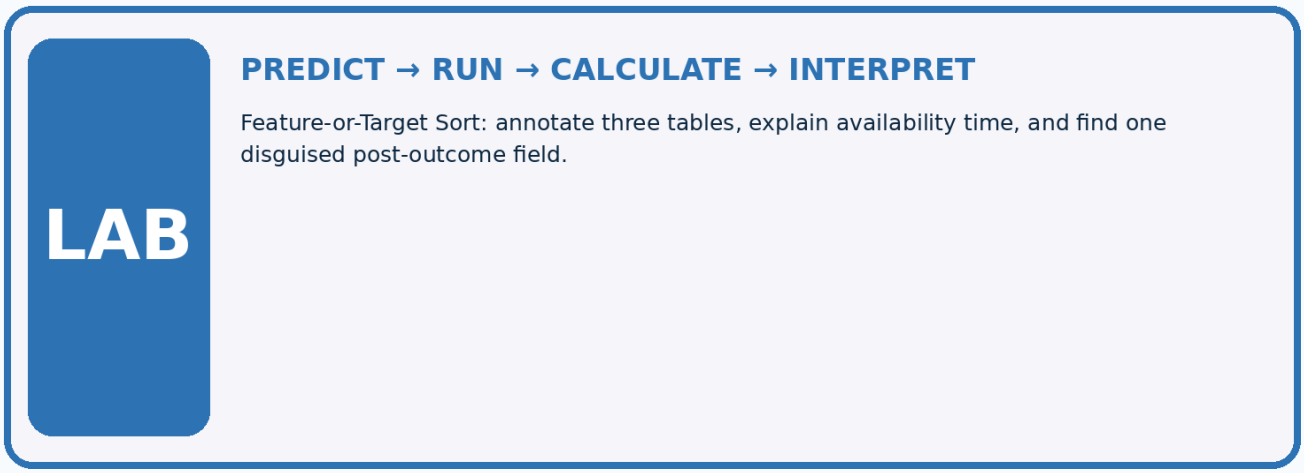
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. **Feature**-or-**Target** Sort: annotate three tables, explain availability time, and find one disguised post-outcome field.
</details>

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: audit the new dispatch-time task

> Explain the timing reason for every included and excluded field.

Independent schema, records and scope

A fictional robot predicts at dispatch whether a parcel trip will take more than 10 minutes.

Fields are route_steps and queue_at_dispatch captured at dispatch, trip_id for tracking, arrival_minutes measured after arrival, and slow = 1 when arrival_minutes > 10. Specify **input**s, order, **target** and exclusions.

For a past trip with steps 30, queue 2, arrival 10, give **feature**s and **label**. For a new trip with steps 24 and queue 0, give **feature**s and explain whether slow is known.

Use the dispatch-time task’s own field names, order and strict more-than boundary. Keep observed completed-trip labels separate from a new trip’s unknown outcome. No future arrival measurement exists for the new trip, and no model prediction is supplied.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Audit a new task independently

Week 2 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A row contains leaf_length_cm=8, leaf_width_cm=3 and species=fern. For predicting species, identify two **feature**s and the **target**.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Define the Plant Classifier **target** and a first **feature** list; record units, allowed values, collection method, and availability time.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A table has leaf length, edge type and verified species. A model predicts species using all three columns. Identify leakage and a legitimate **input** set.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project’s prediction-time contract explicit

> A historical column list is not a feature specification.

This week’s contribution

For the Responsible Plant Classifier, record each feature’s name, units, capture time, source/version, allowed order and missing-data policy. Keep the observed target and its recording rule separate. Candidate features still need a valid predictive-value comparison; availability does not establish usefulness or cause.

Preserve identifiers for tracking or grouped evaluation without silently treating them as predictors.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Retain checkpoint, boundaries and human decision

THIS WEEK'S CHECKPOINT Define the Plant Classifier **target** and a first **feature** list; record units, allowed values, collection method, and availability time.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: name the prediction time, feature snapshot and later target

> A stored value can be real yet unavailable for the intended prediction.

Explain without notes

- Close your notes. Explain this goal in your own words: Separate **prediction input**s from observed **target**s

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
feature	input variable used for prediction	
target	quantity or category to estimate	
label	observed target value in a dataset	
row	one example with fields	

Closer-explanation terms

Word	Meaning
Feature contract	A specification of input meaning units order and timing
Target	The outcome a prediction aims to estimate
Prediction time	The moment when the prediction must be made

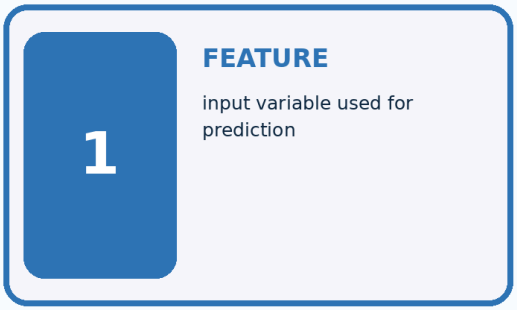
Word	Meaning
Target leakage	Predictor information that reveals the outcome but is unavailable at prediction time

▼ Earlier wording, alternatives and picture transcripts

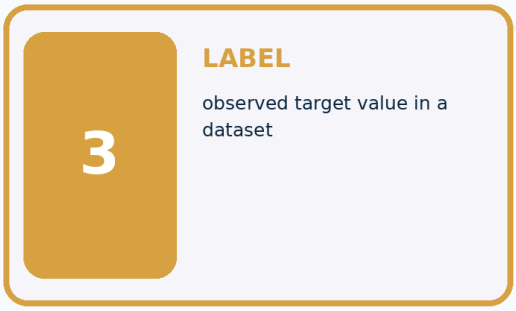
VOCABULARY

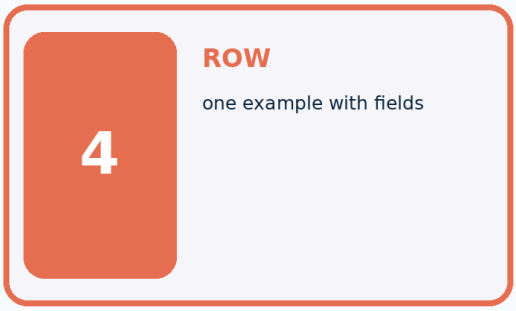
Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR Any column correlated with the answer is automatically a good **feature**.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **feature**, **target**, **label**, row.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE Any column correlated with the answer is automatically a good **feature**.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

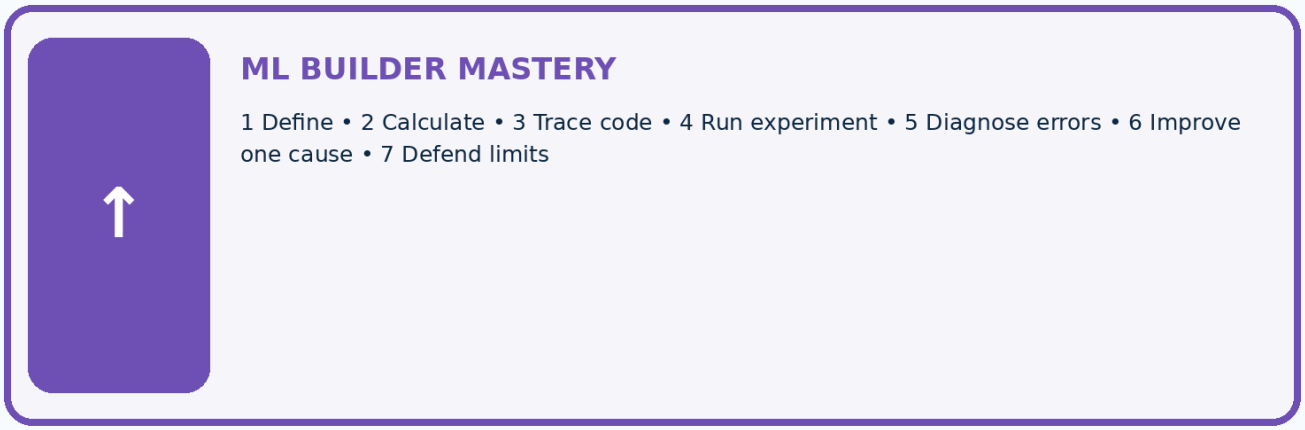
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep Audit the new dispatch-time schema independently for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, **feature/target** roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

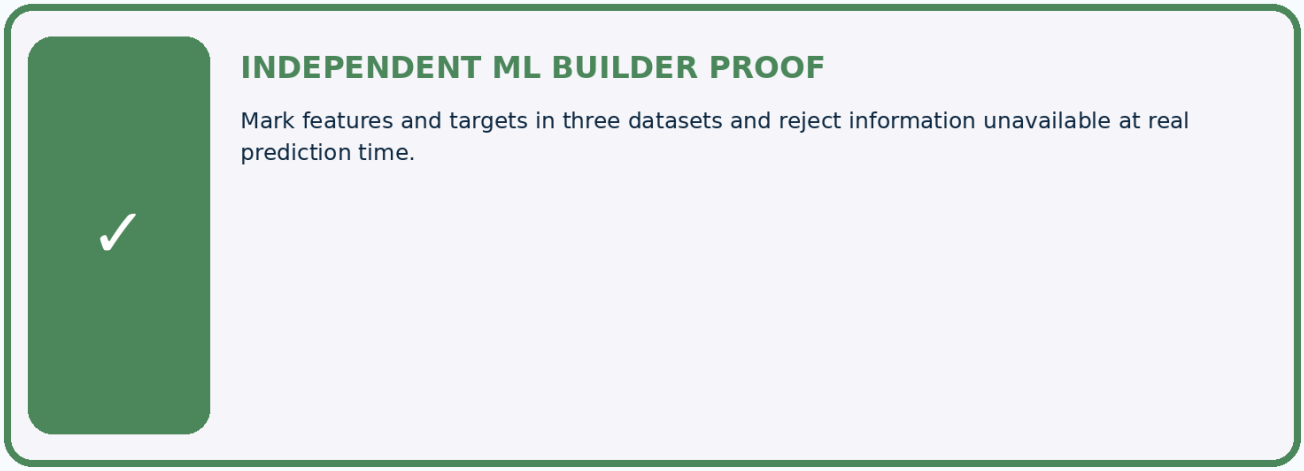
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Mark **feature**s and **target**s in three datasets and reject information unavailable at real **prediction time**.

NEXT BRIDGE Carry the experiment **evidence** into: Classification or Regression?.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Rows contain leaf length, edge type and healthy/spotted **label**.

Identify **feature**s, **target** and **prediction**-time **input**s.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Length and edge type are candidate **feature**s; the **label** is the **target**. Do not supply the true **target** as an **input** to predicting it. Evaluate **feature** usefulness.

Evidence or stage	Value or result
Length	Candidate input feature
Edge type	Candidate input feature
Healthy or spotted	Target label
Prediction time	True target is unavailable

▼ Earlier wording, alternatives and picture transcripts

Week 2 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Separate features from the target

Rows contain leaf length, edge type and healthy/spotted **label**.

Identify **feature**s, **target** and **prediction**-time **input**s.

Length and edge type are candidate **feature**s; the **label** is the **target**. Do not supply the true **target** as an **input** to predicting it. Evaluate **feature** usefulness.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Separate prediction inputs from observed targets	Explain the deciding step, show a trace or test, and state one limit.
Check whether each feature exists at prediction time	Explain the deciding step, show a trace or test, and state one limit.
Repair target leakage and state a feature contract	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Choose **feature**s that exist when a plant **prediction** must be made.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Choose **feature**s that exist when a plant **prediction** must be made.

- Which **label**led step matches this detail? next_week_wilt supplies the **training label**, not an **input** available today.

- 1. Select **input**s

- 2. Mark **prediction time**

- 3. Keep **target** separate

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Keep **target** separate next_week_wilt supplies the **training label**, not an **input** available today.

Read the picture in order. next_week_wilt supplies the **training label**, not an **input** available today. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: A table adds tomorrow_soil_moisture as an **input**. Which conclusion is supported by the supplied facts?

- 1. Reject it for this **prediction**-time contract.

- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Reject it for this **prediction**-time contract. Tomorrow's measurement is unavailable when today's **prediction** must be issued.

Repair and keep the record

Review The ML Lifecycle (ai-4/1). Goal: Distinguish **fitting** a rule from applying a fixed rule. Short practice: Should the team silently retune on the changed test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, **label** your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

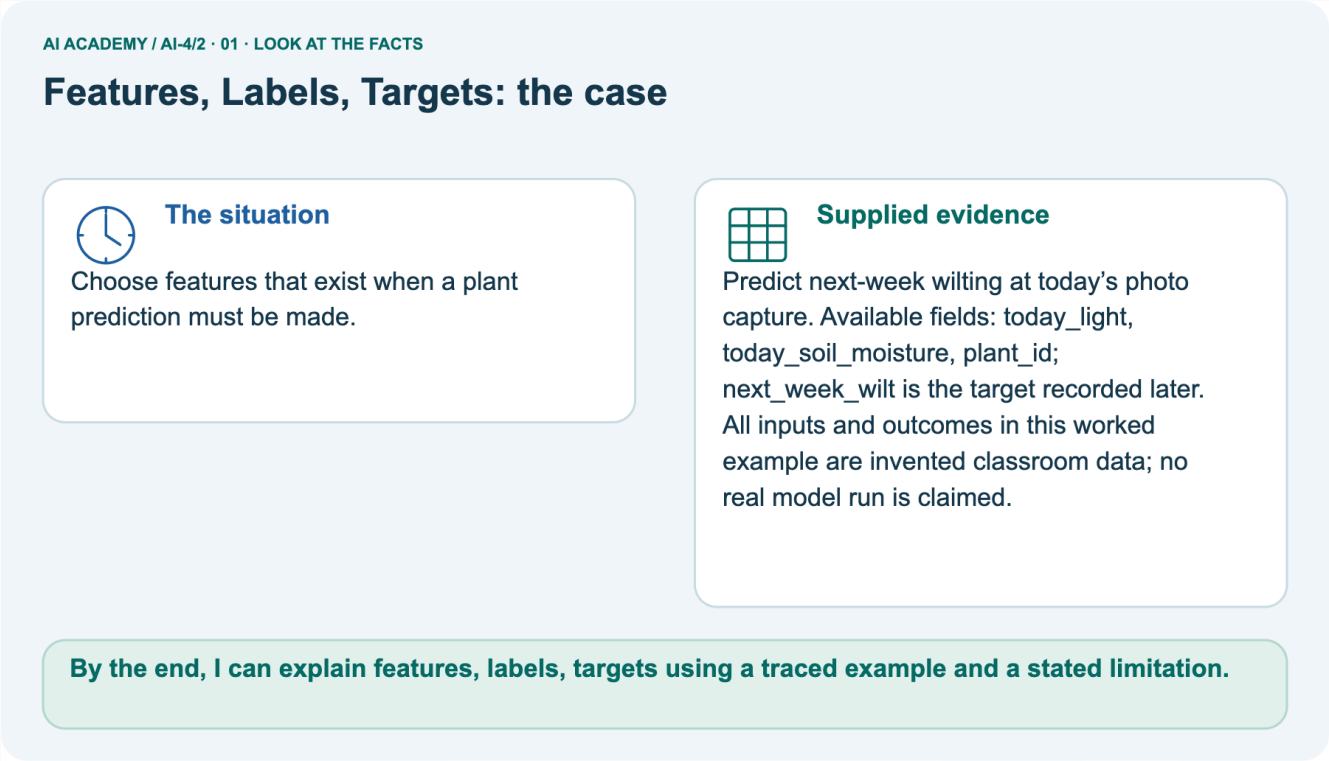
▼ Reference · The six-picture journey

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Features, Labels, Targets: the case



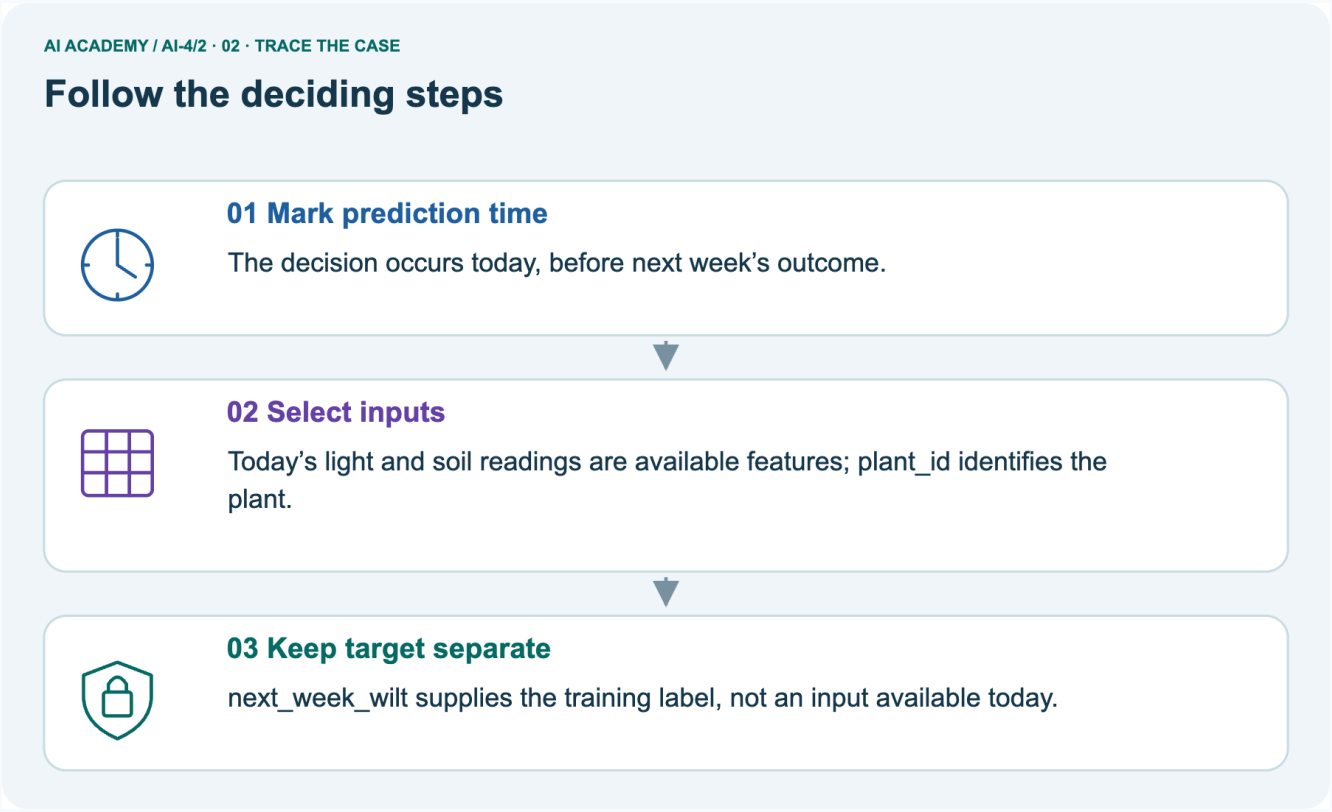
Follow the labelled picture with the matching explanation. Open the vector diagram

Choose **feature**s that exist when a plant **prediction** must be made. Supplied facts: Predict next-week wilting at today's photo capture.

Available fields: today_light, today_soil_moisture, plant_id; next_week_wilt is the **target** recorded later. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Mark **prediction time**: The decision occurs today, before next week's outcome. → Select **input**s: Today's light and soil readings are available **feature**s; plant_id identifies the plant. → Keep **target** separate: next_week_wilt supplies the **training label**, not an **input** available today.

Point to each numbered stage and explain which recorded **input** it uses.

03. Follow information through time

AI ACADEMY / AI-4/2 · 03 · INSPECT THE LESSON MODEL

Follow information through time

Moment	Available information	Allowed use
Checkout	Pages and saved queue	Prediction inputs
Later return	Elapsed days	Create observed label
Historical training	Inputs plus separate label	Fit a model later

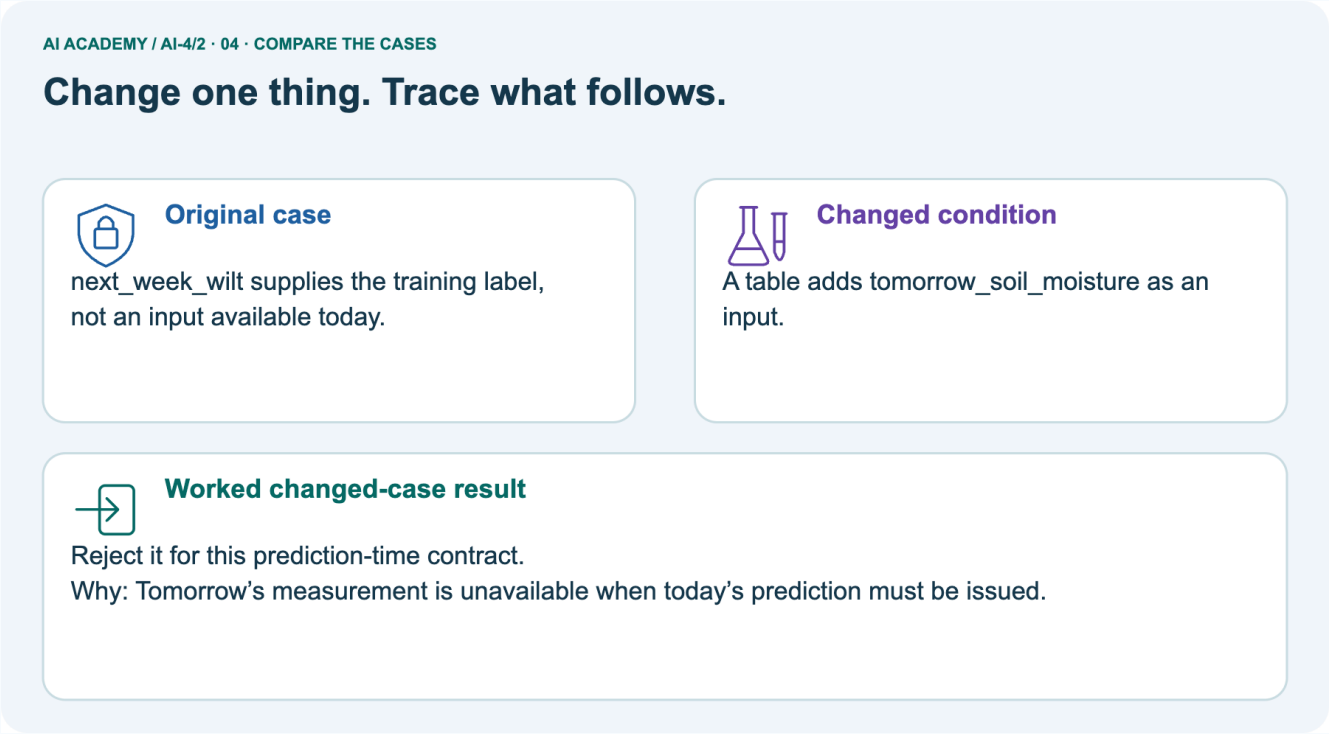
The outcome may teach the model during training, but it cannot be an input to an earlier checkout prediction.

Follow the labelled picture with the matching explanation. Open the vector diagram

Follow information through time. Checkout | Pages and saved queue | **Prediction input**s. Later return | Elapsed days | Create observed **label**. Historical **training** | **Input**s plus separate **label** | Fit a model later

Read the **label**s and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



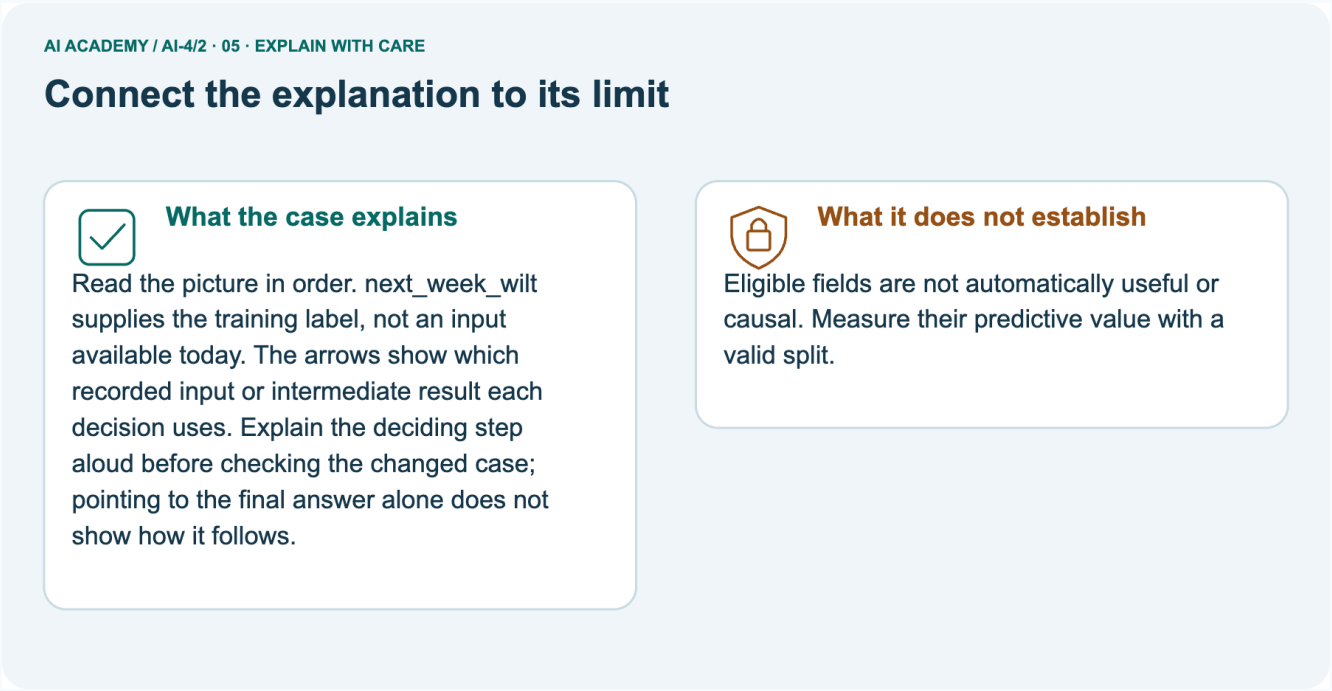
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: next_week_wilt supplies the **training label**, not an **input** available today. Changed condition: A table adds tomorrow_soil_moisture as an **input**. Expected result: Reject it for this **prediction**-time contract.

Reason: Tomorrow's measurement is unavailable when today's **prediction** must be issued.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

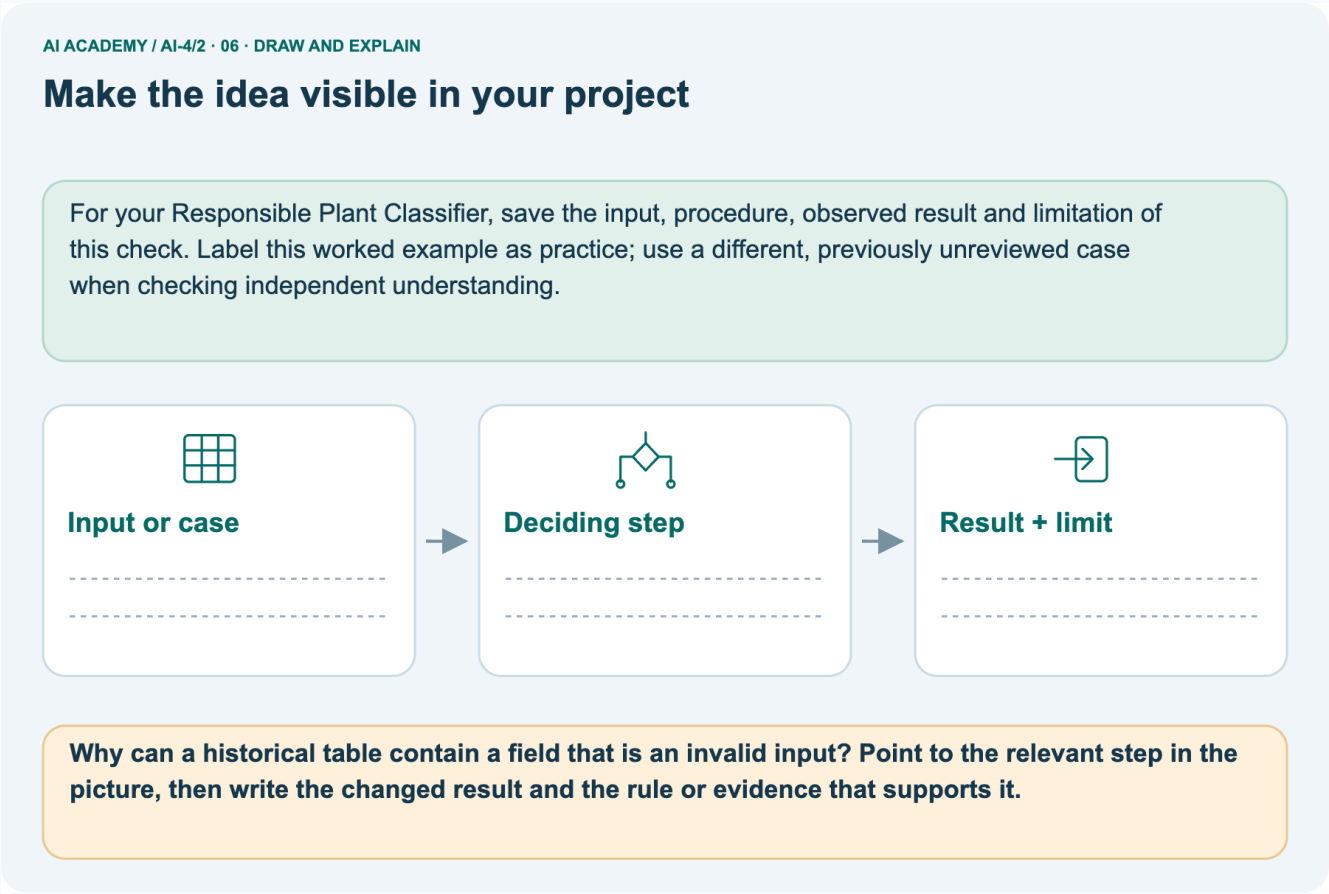
Read the picture in order. next_week_wilt supplies the **training label**, not an **input** available today. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Eligible fields are not automatically useful or causal.

Measure their predictive value with a valid split.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why can a historical table contain a field that is an invalid **input**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Features, Labels, Targets: the case

02. Follow the deciding steps

03. Follow information through time

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.